- The acceleration due to gravity is denoted by $g\,\text{m}\,\text{s}^{-2}$. Unless otherwise instructed, when a numerical value is needed, use $g = 9.8$.

13 A 15 kg box is suspended in the air by a rope which makes an angle of 30° with the vertical. The box is held in place by a string which is horizontal.

(a) Draw a diagram showing the forces acting on the box. **[1]**

(b) Calculate the tension in the rope. **[2]**

(c) Calculate the tension in the string. **[2]**